

Analysis of bulk RNA-seq data
Analysis of Next-Generation Sequencing Data

Fundamental concepts of RNA sequencing experiments

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Applied Bioinformatics Core

Slides at <https://bit.ly/2T3sjRg>

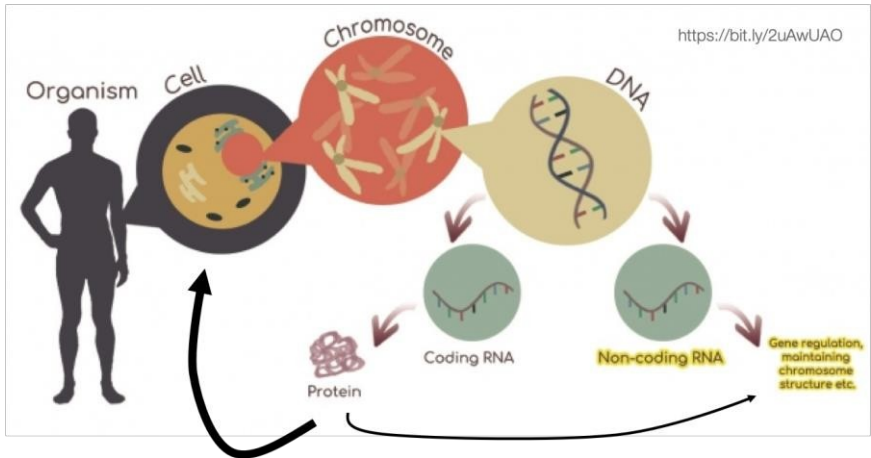
February 18, 2020



¹https://physiology.med.cornell.edu/faculty/skrabanek/lab/angsd/schedule_2020/

- 1 Why study RNA?
- 2 Different types of RNA – different library preps
- 3 Gene expression : Microarray vs RNA-seq

Why study RNA?



DNA is just the blueprint, it is not an effector molecule.

DNA is just the blueprint, it is not an effector molecule

GENOMICS

- DNA sequence of an organism
- genetic basis of phenotypic differences
- sites of DNA-protein or DNA-RNA interactions
- sites of open vs closed chromatin

TRANSCRIPTOMICS

- = characterization of gene products
- identification of specific RNAs
- quantification of RNAs
- RNA-protein interactions
- RNA structure

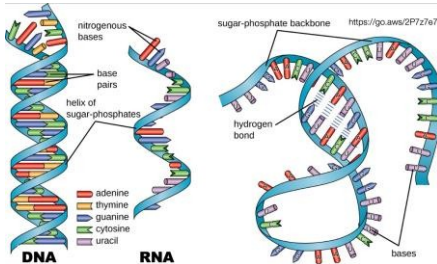
In order to understand the functional consequences of a DNA sequence, we need to study its products, i.e.
RNA and proteins.

Different types of RNA – different library preps

DNA and RNA have different properties

DNA

- usually double-stranded
- very stable
- mutations are heritable
- same amount in (almost) all cells
- same sequence in every cell of an organism

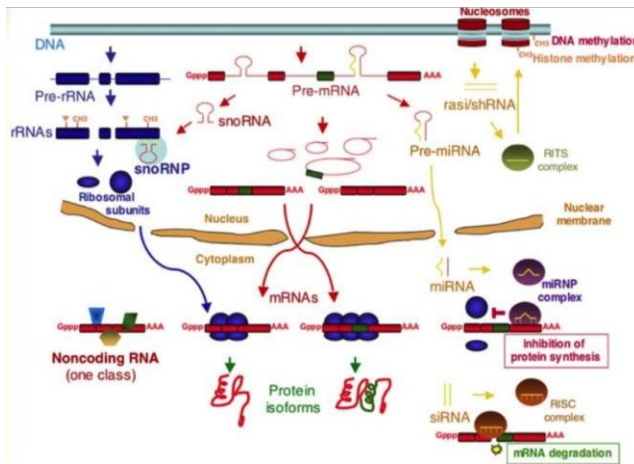


RNA

- generally single-stranded, but with the capacity for complementary base pairing $\Rightarrow$ ability to form myriad different shapes
- usually fairly short-lived (minutes to hours)
- easily degraded/damaged without protection
- mutations are not passed on
- individual transcript *amounts* differ greatly depending on the gene, the cell type, the developmental status, the environment etc.
- transcript *sizes* range from 10-20bp to several kb

Different types of RNA

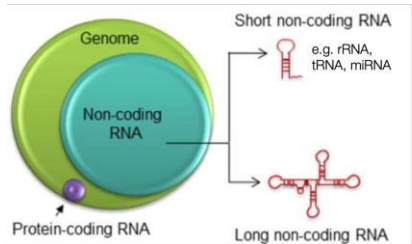
There are numerous different types of **functional** RNA molecules *in addition* to messenger RNA, which does *not* carry out a function of its own except transporting the DNA code (genetic information) into the cytoplasm where it can be translated into proteins.



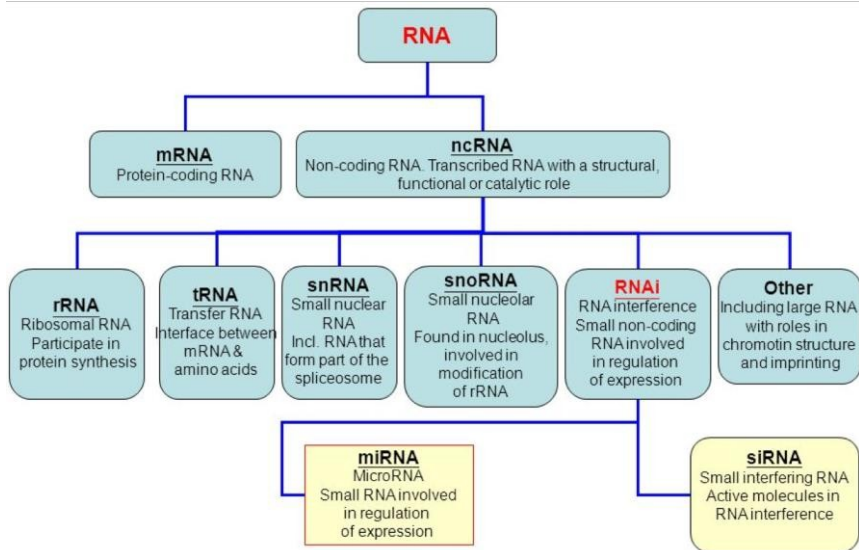
Different types of RNA

- Ca. 75% of the human genome can be transcribed(= copied into RNA) but <3% of the genome is subsequently translated into proteins genes can therefore be
 - coding** (final product: protein) or
 - non-coding** (final product: RNA)
- Non-coding RNAs cover a wide range of functions including (⇒ ribosomal RNA, transfer RNA) and gene expression regulation

See [Wilkes et al. \[2017\]](#), [Bartoszewski and Sikorski \[2018\]](#) and [Dai et al. \[2020\]](#) for an introduction into the diverse RNA families and their functions.



Different types of RNA (there are more!)



Typical applications of RNA-seq

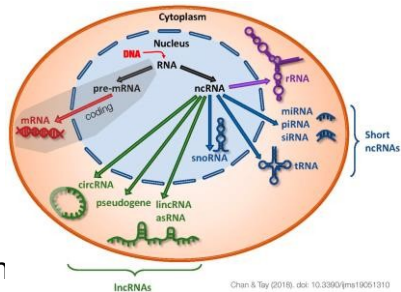
- **Identification** of transcripts – *which portions of the genome are expressed?*
 - ▶ identification of splice variants
 - ▶ transcriptome assembly
 - ▶ detection of gene fusion events
- **Quantification** of transcripts
 - ▶ comparison of different cell types/conditions/diseases and their effect on individual mRNA quantities
 - ▶ allele-specific expression

Illumina technology is best suited for the **quantification of known** transcripts; its short reads are not a good match for the identification of novel transcripts in very complex transcriptomes such as the ones found in mammals.

Sequencing prep protocol depends on the RNA pro

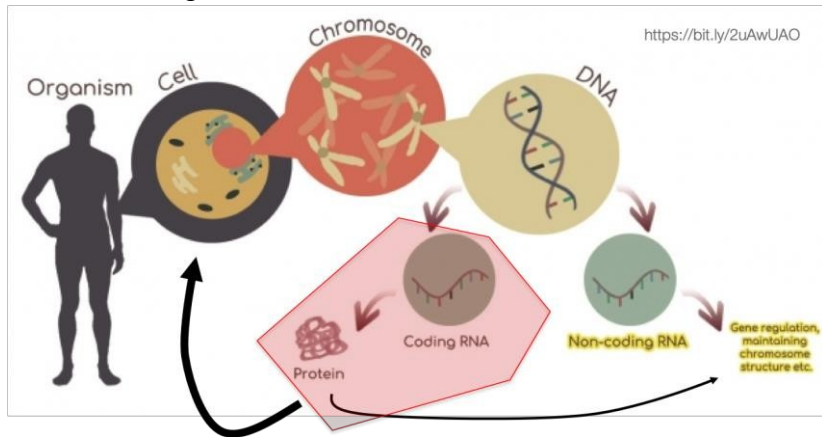
It is not a one-size-fits-all situation!

- abundance and stability
 - rRNA: 90-95% (!)
 - tRNA: 3-5%
 - mRNA: 2%
 - all other non-coding RNAs: well below 1%
- cellular location
 - most are in the cytoplasm
- size
 - miRNAs 18-23bp
 - mRNA: several 100 to 1000 bp
- specific sequences/modification
 - poly(A) tails of mRNA
 - 2D structure
 - antisense transcripts

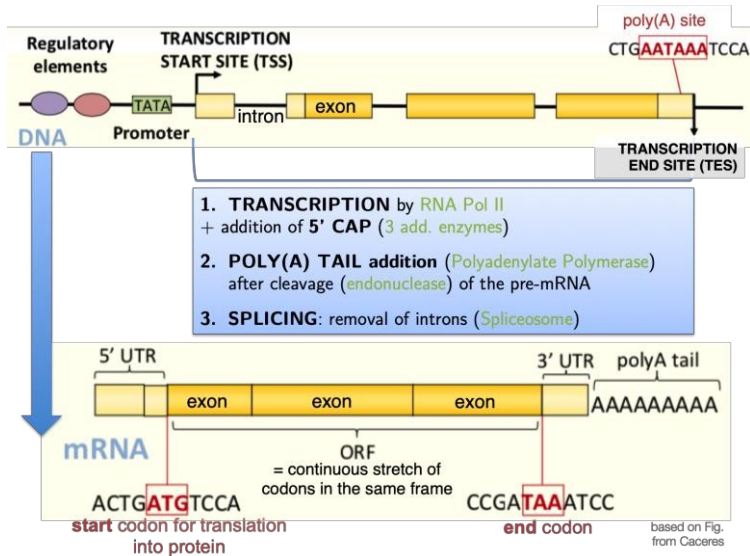


Focus today messenger RNA

mRNA amounts are used as a proxy for the amounts of their corresponding proteins within a given tissue.



Focus today messenger RNA



How to profile gene expression?

How to profile gene expression?

1. Quantitative Reverse Transcription and Polymerase Chain Reaction (qRT/PCR)
2. Northern Blotting
3. Fluorescence *In situ* Hybridization (FISH)
4. Microarrays
5. RNA-seq

DATA GENERATION



Convert to ds cDNA

Amplify, label & fragment

Hybridize to array

Scan array

RNA extraction

RNA quantification & quality assessment



PolyA selection or rRNA depletion

Fragment

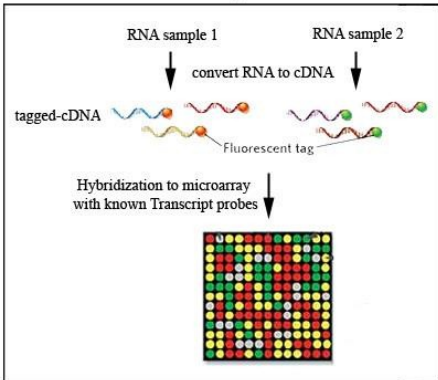
Convert to cDNA & ligate adapters

Amplify

Sequence

Microarray vs RNA-seq

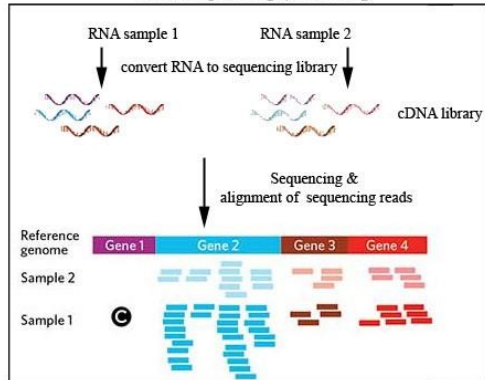
Microarray



relative intensity
=
expression levels

Low sensitivity
Low dynamic range
known transcript only
No alternative splicing information
lower cost

RNA Sequencing (RNA-Seq)



High sensitivity
High dynamic range
Novel transcripts sequences identified
structural variation & alternative splicing revealed
unlimited sample comparisons

Sequencing Reads
=
expression levels

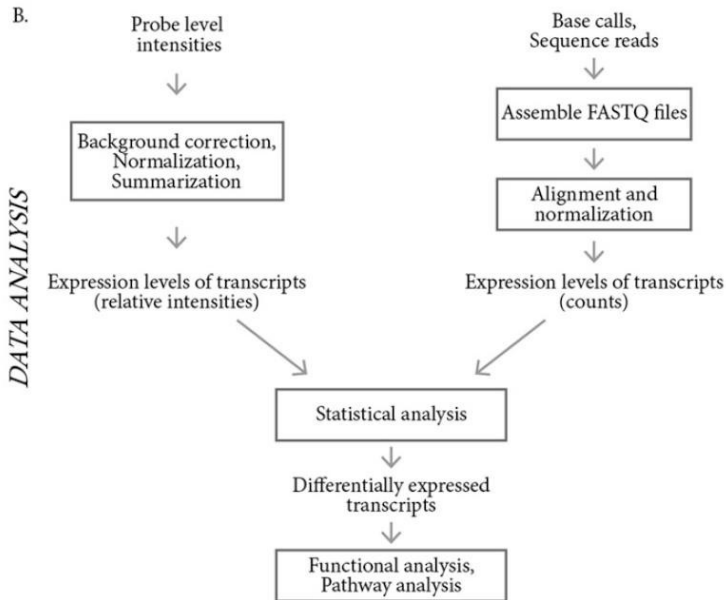


Fig. 1 Workflow of typical gene expression profiling with expression array or RNA-Seq

Table 1
Comparisons of RNA-Seq and microarray in the literature—technology and design details

Reference	Array type ^a	RNA-Seq platform	Library prep	Sequencing details ^b	Sample	Experimental design
Marioni et al. [41]	Affymetrix HG-U133 Plus 2	Illumina Genome Analyzer	polyA+ selection	32 bp, ~8–15mil	One human male sample from liver and kidney	Seven technical replicates, two cDNA conc., two sequencing runs, three arrays
Bottomly et al. [43]	Affymetrix MOE 2.0, Illumina MouseRef-8	Illumina GAIIX	polyA+ selection, Illumina mRNA-Seq Sample Preparation kit	300 bp, 21 samples in 21 lanes	Adult B6 and D2 strain mice; variation in prep dates, sex of samples among platforms	10+ strain replicates, all male for RNA-Seq and Illumina array, equal male/female Affymetrix array
Raghavachari et al. [42]	Affymetrix Human Exon 1.0 ST	Illumina GAIIX	polyA+ selection (Illumina recommendations)	Single lane, 36 cycles, sequencing depth ~10mil	Human Whole Blood (PAXGene), GLOBINclear depletion	Six patients sickle cell disease, four controls
Zwemer et al. [44]	Affymetrix HG-U133 Plus 2	Illumina HiSeq 2000	NuGEN Ovation RNA-Seq V2 (Hudson Alpha Institute)	Paired end, 50 bp, ~17–100 mil pairs; low alignment stats (5–35% for all but 1)	Human amniotic fluid	Three male, two female fetuses

SEQC Consortium [45]	Affymetrix HG-U133 Plus 2, Affymetrix HuGene2.0, Affymetrix PrimeView, Illumina Bead array	Illumina HiSeq 2000, Life Technologies SOLiD 5500, Roche 454 GS FLX	Varies with site	Paired end; Illumina: 100 bp, ~110 mil pairs; Solid: 51/36 bp, ~50mil; Roche: ~1mil	Human, rat, multiple tissue types	Many samples in consortium
Zhao et al. [26]	Affymetrix HG-U133 Plus 2	Illumina HiSeq 2000 (BGI)	polyA+ selection	Paired end, 90 bp	Human CCR6+ CD4+ T cells	6 time points, +/- stimulation in duplicate
Yu et al. [27]	Agilent Hum Genome 4x44K, Illumina Hum HT12v4, Affymetrix Hum Gene 1.0 and HTA2.0	Illumina HiSeq 2500	SMARTer UltrLow RNA kit, Epicentre RiboZero	50 bp, ~36mil reads	One pool of human bone marrow RNAs, Universal Human Ref RNA (Agilent)	Two samples plus ratio-varied mix of the two samples (1-3 reps)
Nazarov et al. [46]	Affymetrix HTA 2.0	Illumina HiSeq 2000	Ribosomal RNA depletion	Paired end, 77 bp, 120-280 mil reads	Matched primary human lung tumor and adjacent tissue	Nine matched samples

^aTotal RNA

^bUnstranded and single end unless otherwise stated

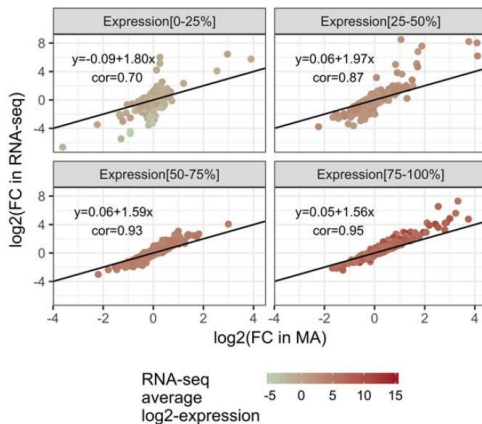


Fig. 2 Log2-Fold change (Condition 2 vs 1) from microarray and RNA-Seq data. Each point represents a gene that is represented on both platforms (8520 genes total). Genes were divided into four panels based on quantiles of average RNA-Seq log2-expression. Darker shades represent larger average abundance. Pearson's correlation as well as the intercept and slope from the univariate regression model of log2-FC in RNA-Seq regressed on log2-FC in array are displayed. Black lines represent the diagonal slope of 1, which would occur when fold changes in the platforms are identical. A slope larger than 1 from the univariate regression denote a trend of larger fold change magnitudes measured in RNA-Seq

Table 2

Number of genes and gene ontology terms detected by each platform separately as well as with both platforms

Number	Microarray	RNA-Seq	Platform overlap
Genes ^a quantified: Protein-coding (All RNA) genes	26,596 (65,956)	22,001 (45,706)	22,001 (37,921)
Genes ^a after low expression filtering ^b	9262 (13230)	14,554 (16696)	8254 (8520)
DE genes ^a FDR 5%; <i>As subset of overlap^c</i>	671 (962); <i>601 (604)</i>	1918 (2095); <i>1241 (1268)</i>	<i>550 (553)</i>
DE genes ^a FDR 5% and log2FC > 0.5; <i>As subset of overlap^c</i>	484 (745); <i>431 (434)</i>	1595 (1770); <i>956 (983)</i>	<i>411 (414)</i>
GO terms identified	11,365	12,964	11,298
GO terms $p < 0.05$	889	1682	662

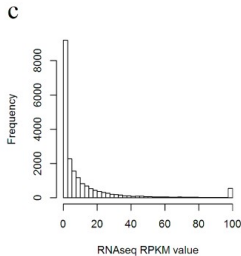
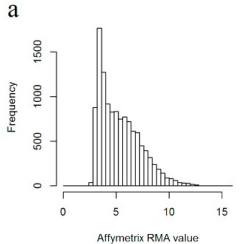
^aNumber of protein coding genes (number of total genes)

^bFiltering: For microarray data, the 95th percentile of the normalized intensity distribution of antigenomic background control transcript clusters was calculated as a background cutoff, and transcript clusters were removed from further analysis if fewer than two samples exhibited normalized expression levels above this cutoff. RNA-Seq genes were removed from further analysis if fewer than two samples exhibited CPM greater than 0.2. This constitutes the set of genes tested for differential expression

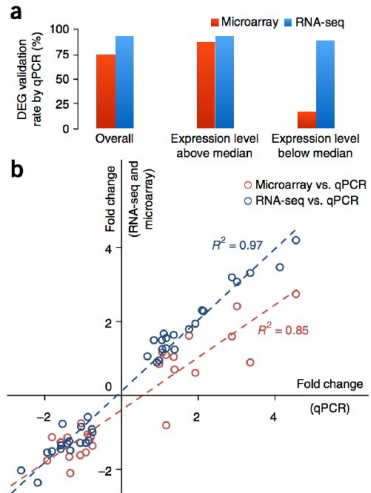
^cNumbers in italics are a subset of the 8520 genes found on both platforms. For each platform, these correspond to significant detection in one platform regardless of whether it was significant for the other platform

^dDE denotes differentially expressed genes, GO denotes Genome Ontology terms belonging to the Biological Process ontology

Microarray vs RNA-seq



Guo et al. (2013) *Plos One*



Wang et al (2014) *Nature Biotech.*

How to choose gene expression assay?

	<i>Value</i>	<i>Pros</i>	<i>Cons</i>
<i>Arrays</i>	- Accurate and reproducible expression data - Established, reliable platform	- Standardized workflows and user-friendly software available for data processing and analysis - Fast turnaround times, esp. for small studies	- Limited dynamic range for fold change measurements - Require known sequence info for array fabrication - Limited detection of very low abundance transcripts
<i>RNA-Seq</i>	- Accurate and reproducible expression data - Transcript sequence information included	- Extended dynamic range - Ability to detect rare transcripts with deep sequencing - Ability to identify novel transcripts and sequence variation	- Complex data analysis w/ no 'gold-standard' pipeline - Requires mRNA selection or abundant transcript removal to avoid deep sequencing costs

Fig. 3 Performance and use considerations for gene expression profiling with microarrays versus RNA-Seq

DECIDING BETWEEN EXPRESSION ARRAYS AND RNA-SEQ

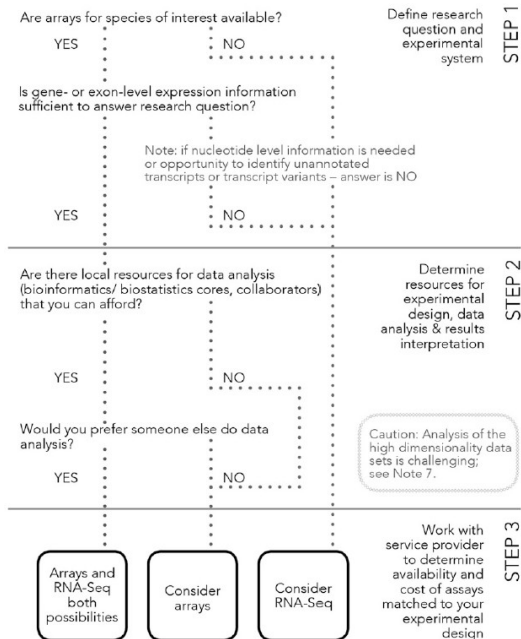
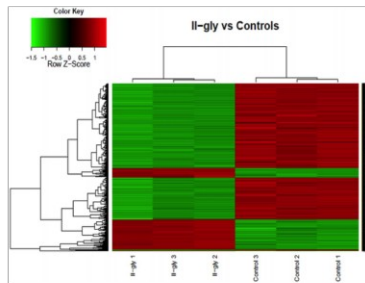
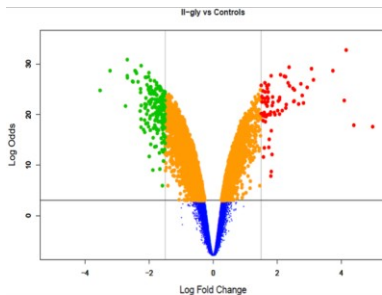


Fig. 4 Technology platform selection process for gene expression profiling

Focus today messenger RNA

Bulk RNA-seq of mRNA

- expression quantification of (mostly) mRNA transcripts
- extracted from populations of cells
- and tested for gene-specific differences between distinct conditions

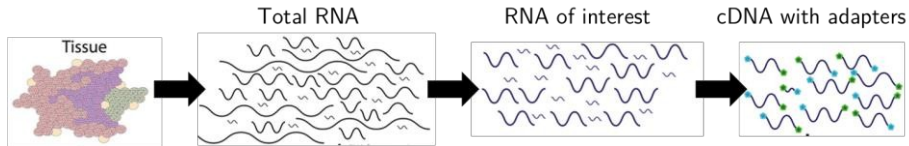


Typical questions addressed with bulk RNA-seq

- Does a certain treatment induce gene expression changes? And if it does, which genes are most strongly affected?
- How does the gene expression profile of a cancer cell differ from a healthy cell?
- Which genes are turned on/off during the course of embryonic development?
- Which genes differ in mice that have been engineered to lack a certain gene? E.g., which genes – in addition to the one that's been “knocked-out” – may be depleted or overcompensating for the loss?
- Which genes are activated in response to an environmental stimulus e.g. heat shock or alcohol poisoning?
- How does the gene expression profile change in the same tissue in an aging individual?
- . . .

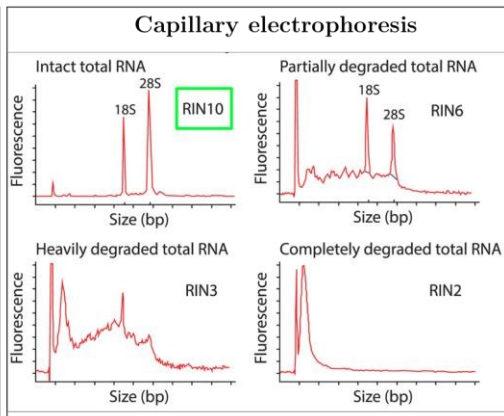
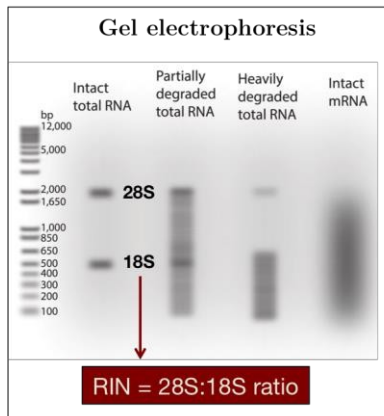
General steps of RNA-seq preparation

- 1 RNA **extraction**² (cell lysis, RNA purification)
- 2 **enrichment** of the RNA of interest
- 3 **fragmentation** (200 bp)
- 4 **cDNA** synthesis
- 5 library prep to obtain cDNA with **adapters** for sequencing



²Most standard extraction methods will lose RNA <100 bp!

QC of RNA extraction

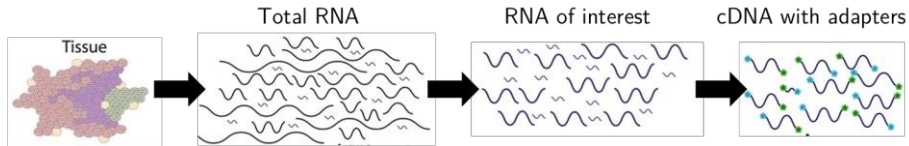


Griffith et al. (2015). doi: 10.1371/journal.pcbi.1004393

Avoid degraded RNA! Optimum: RNA Integrity Score (RIN) of 10.

General steps of RNA-seq preparation

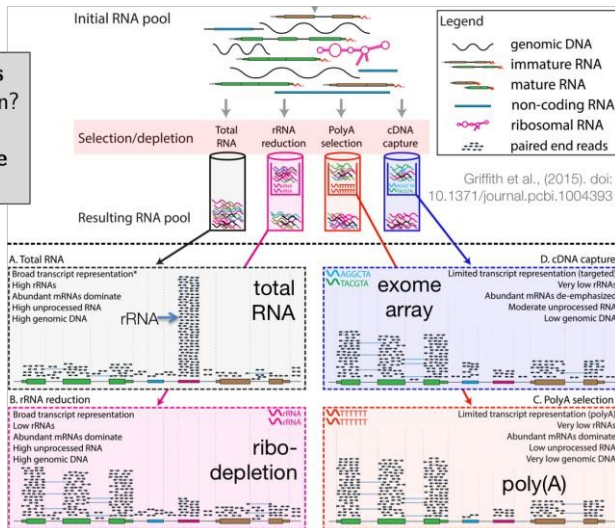
- 1 RNA extraction (yes, RNA purification)
- 2 **enrichment of the RNA of interest**
 - ▶ mRNA: poly(A) enrichment vs. ribosomal-depletion
 - ▶ small RNAs: size-based enrichment
- 3 fragmentation (200 bp)
- 4 cDNA synthesis
- 5 library prep to obtain cDNA with adapters for sequencing



Every step has consequences – example: enrichment strategies

which **transcripts**
are you interested in?

what type of **noise**
can you tolerate?



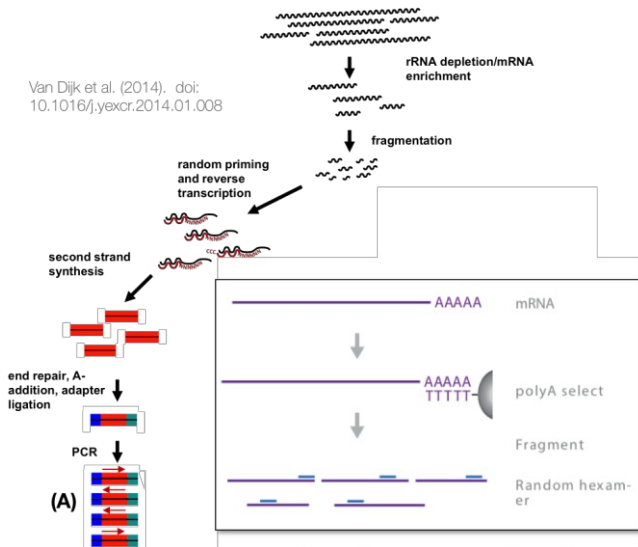
■ rRNA

■ protein coding
(strongly expressed)

■ protein coding
(lowly expressed)

The most common library preparation methods

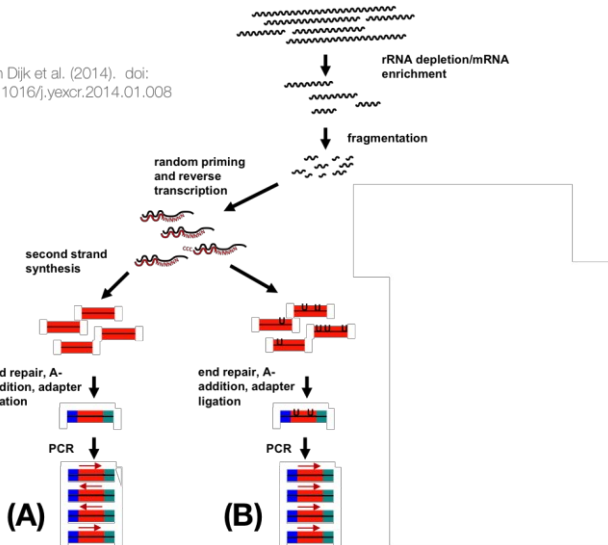
Van Dijk et al. (2014). doi:
10.1016/j.yexcr.2014.01.008



- **(A)** classical unstranded mRNA library prep

The most common library preparation methods

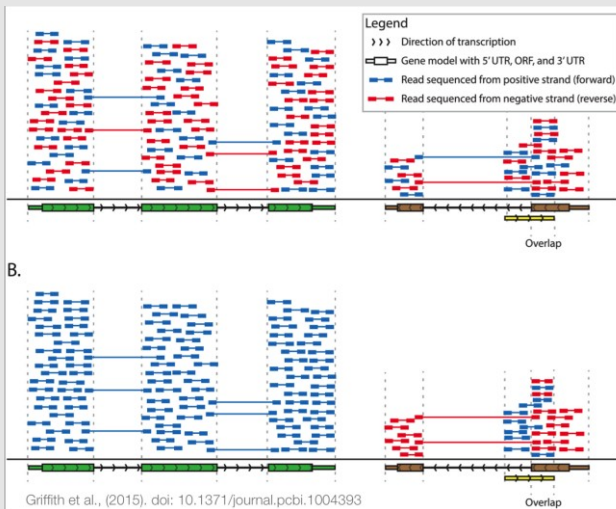
Van Dijk et al. (2014). doi:
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- (A) classical unstranded mRNA library prep
- (B) stranded mRNA (dUTP-based) (see [Levin et al. \[2010\]](#) and [Zhao et al. \[2015\]](#) for details)

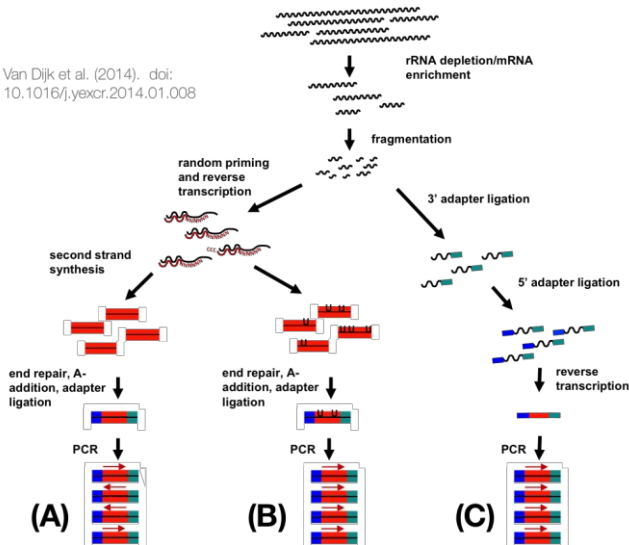
The most common library preparation methods

Unstranded vs. stranded



The most common library preparation methods

Van Dijk et al. (2014). doi:
10.1016/j.yexcr.2014.01.008



- (A) classical unstranded mRNA library prep
- (B) stranded mRNA (dUTP-based) (see [Levin et al. \[2010\]](#) and [Zhao et al. \[2015\]](#) for details)
- (C) small RNAs (miRNA, piRNA, tRNA, ... <100 bp) using 2 adapters – not optimal for differential expression analyses!

Every step has consequences

- Do not mix different strategies for samples that are to be compared to each other!
 - ▶ extraction, enrichment, library prep

There are many papers comparing different aspects of different RNA-seq approaches, e.g.

- Library preparation methods for next-generation sequencing: Tone down the bias* [van Dijk et al., 2014]
- Systematic comparison of small RNA library preparation protocols for next-generation sequencing* [Dard-Dascot et al., 2018]
- A comprehensive assessment of RNA-seq protocols for degraded and low-quantity samples.* [Schuijjer et al., 2017]
- many more – PubMed is your friend!

Make an informed decision!